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Systems and methods for assembling stiffened composite structures

Abstract

Systems (**100**) for assembling stiffened composite structures comprise a layup mandrel (**102**) and a network (**104**) of fluid lines (**116**) configured to be selectively and fluidically coupled to string bladders (**118**) positioned relative to the layup mandrel (**102**).

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Background/Summary

FIELD

(1) The present disclosure relates to systems and methods of assembling stiffened composite structures.

BACKGROUND

(2) Stiffened composite structures are structures that are constructed of composite materials, such as fiber reinforced composite materials, and typically include some form of structural frame that carries a skin. Some modern aircraft fuselages are examples of stiffened composite structures that include a skin operatively coupled to frame members that extend circumferentially around and that are spaced longitudinally along the inside of the fuselage and stringers that extend longitudinally along the inside of the fuselage and that are spaced circumferentially around the inside of the fuselage. Stiffened composite fuselages may be constructed utilizing an inner mold line layup mandrel that includes stringer cavities, or forms, that extend longitudinally along the mandrel. Typical stringers, such as hat-shaped stringers, define cavities themselves, with these cavities needing to be filled with a structure that will result in a flush surface to which the skin may be applied. These filler structures often are referred to as stringer mandrels or stringer bladders. Some stringer bladders are solid and made of a rubber or similar material. Other stringer bladders are hollow. Once the skin is applied over the stringers and the stringer bladders on the inner mold line layup mandrel, the stringers and skin are vacuum-compacted and co-cured in an autoclave. The vacuum compacting process utilizes a flexible sheet of material and sometimes caul plates that collectively extend across the stringers and skin and are sealed to the inner mold line layup mandrel around the outer perimeter of the stringers and skin. A vacuum is then applied between the sheet of material and the inner mold line layup mandrel to compact the stringers and skin together. When hollow stringer bladders are used, the internal volumes of the stringer bladders must be isolated

from the vacuum and fluidically open to atmosphere to avoid the string bladders collapsing during the vacuum compacting process. This vacuum compacting process is referred to in the aerospace industry as “bagging” and is labor and time intensive. In particular, the integrity of the seal between the flexible sheet of material and the inner mold line layup mandrel is critical. Conventionally, the sheet of material, or “bag,” is perforated by the vent lines that are connected to the internal volumes of the stringer bladders, resulting in numerous potential leak points.

SUMMARY

(3) Systems for assembling stiffened composite structures comprise a layup mandrel and a network of fluid lines configured to be selectively and fluidically coupled to stringer bladders relative to the layup mandrel. In some examples, the network of fluid lines extends through the layup mandrel. Some systems further comprise one or more covers configured to be operatively coupled to the layup mandrel to define a passage extending along the layup mandrel, such that the network of fluid lines is positioned at least partially within the passage when the one or more covers are operatively coupled to the layup mandrel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an example of a stiffened composite structure in the form of an aircraft.
- (2) FIG. 2 is an isometric view of an example of a stiffened composite structure in the form of a barrel section of an aircraft.
- (3) FIG. 3 is a representation of an example inner mold line layup mandrel for assembling a barrel section of an aircraft.
- (4) FIG. 4 is a schematic diagram representing systems for assembling stiffened composite structures.
- (5) FIG. 5 is an isometric view of a portion of an example of a system for assembling stiffened composite structures, shown in a cure configuration.
- (6) FIG. 6 is another isometric view of the portion of the example system of FIG. 5, shown in a bladder leak test configuration.
- (7) FIG. 7 is another isometric view of a portion of the example system of FIG. 5, shown in a manifold leak test configuration.
- (8) FIG. 8 is another isometric view of a portion of the example system of FIG. 5, shown in a configuration with one stringer and bladder absent.
- (9) FIG. 9 is a flowchart schematically representing methods of assembling stiffened composite structures.

DESCRIPTION

(10) Systems and methods for assembling stiffened composite structures are disclosed herein. In FIG. 1, an example of an apparatus **10** that may be constructed from stiffened composite structures **12** is provided in the form of an aircraft **14**; however, other apparatuses **10** are within the scope of the present disclosure, and the present disclosure is not limited to aircraft and aircraft applications. For example, other apparatuses **10** that may be constructed of stiffened composite structures **12** include (but are not limited to) spacecraft, watercraft, land vehicles, wind turbines, structural towers and masts, etc. Moreover, aircraft **14** may take any suitable form, including commercial aircraft, military aircraft, private aircraft, or any other suitable aircraft. While FIG. 1 illustrates aircraft **14** in the form of a fixed wing aircraft, other types and configurations of aircraft are within the scope of aircraft **14** according to the present disclosure, including (but not limited to) helicopters and other rotorcraft.

(11) Aircraft **14** include a fuselage **16**, which generally corresponds to the main body of the aircraft

14 for holding passengers, crew, cargo, and/or equipment, for example, depending on the particular configuration and/or function of an aircraft **14**. Typically, although not required, the fuselage **16** of an aircraft **14** is elongate and somewhat cylindrical or tubular. In some embodiments, the fuselage **16** may be constructed of multiple sections **18** that are longitudinally spaced along the fuselage **16** and operatively coupled together to define the fuselage **16**. As used herein when referencing an aircraft **14**, a fuselage **16**, and/or a corresponding stiffened composite structure **12**, the longitudinal direction refers to the fore-aft direction, corresponding to a longitudinal, or long, axis, of the fuselage **16**. Additionally, the terms “inner” and “outer” when used in connection with an aircraft **14**, a fuselage **16**, and/or a corresponding stiffened composite structure **12** refer to the radially inward side and the radially outward side, respectively, of the corresponding stiffened composite structure. Accordingly, the outer side of a fuselage **16** or a component part thereof generally faces away from the fuselage **16**, and the inner side of a fuselage **16** or a component part thereof generally faces the internal volume that is defined by the fuselage **16**. Similar relative terms may be used with respect to stiffened composite structures **12** other than fuselages **16** and with respect to apparatuses **10** other than aircraft **14**.

(12) In FIG. **1**, five fuselage sections **18** are indicated schematically, including a nose and tail section; however, any number or size and shape of sections **18** may be used to construct a fuselage **16**. Sections **18** additionally or alternatively may be referred to as barrel sections **18** of a fuselage **16**, with each barrel section **18** generally defining a length of the fuselage **16**. Additionally or alternatively, as schematically illustrated in FIG. **1**, a section **18** may include, or be constructed of, more than one subsection **20**. In the schematically illustrated and optional example of FIG. **1**, a section **18** is shown schematically and optionally to include an upper subsection **20** and a lower subsection **20**; however, a section **18** may be constructed of any suitable number, configuration, and placement of subsections **20**, including no subsections **20** at all.

(13) Aircraft **14** also may include wings **22**, horizontal stabilizers **24**, and a vertical stabilizer **26**, each of which may be constructed as a unitary structure or in subsections that are subsequently assembled together. One or more of a fuselage **16**, a fuselage section **18**, a fuselage subsection **20**, a wing **22**, a horizontal stabilizer **24**, a vertical stabilizer **26**, or a structural subsection thereof may be constructed of and/or may be described as a stiffened composite structure **12** according to the present disclosure.

(14) Stiffened composite structures **12** according to the present disclosure generally encompass structures that are defined by a structural frame **28** and a skin **30** that is operatively coupled to and supported by the structural frame **28**. FIG. **2** depicts an illustrative, non-exclusive example of a stiffened composite structure **12** in the form of a barrel section **18** of a fuselage **16** of an aircraft **14**. The skin **30** may be described as defining the outer shape of the fuselage **16**. As indicated in FIG. **2**, structural frames **28** include a plurality of stiffeners **32** operatively coupled to the inner side of the skin **30**. In the example of fuselage **16**, the stiffeners **32** include a plurality of frame members, or frames, **34** that are operatively coupled to and that extend circumferentially around the inner side of the skin **30** and that are spaced longitudinally along the fuselage **16**. The stiffeners **32** also include a plurality of stringers **36** that are operatively coupled to and that extend longitudinally along the inner side of the skin **30** and that are spaced circumferentially around the fuselage **16**. In some embodiments of stiffened composite structures **12**, although not required, the skin **30** may be constructed of a plurality of skin segments **38** that collectively define the skin **30**, or at least a portion of the skin **30** of a stiffened composite structure **12**.

(15) FIG. **3** schematically illustrates an example of a layup mandrel **102** that may be used to assemble stiffened composite structures **12**. In the illustrated example, the layup mandrel **102** may be described as comprising an outer surface **106** that defines a plurality of spaced-apart stringer forms **108** that extend longitudinally along the outer surface **106**, with each stringer form **108** having an outer surface that corresponds to one side of a respective stringer **36** that is to be loaded into the stiffener form **108**. Each stringer form **108** also has an open end **110**. The outer surface **106**

of the layup mandrel **102** may be described as defining a primary skin contour **148** between the stringer forms **108** and that ultimately shapes the skin **30** of the stiffened composite structure **12** being assembled on the layup mandrel **102**. The schematically represented example layup mandrel **102** of FIG. **3** is configured for assembling a cylindrical or other tubular structure, such as barrel section **18** of a fuselage **16** of an aircraft **14**; however, layup mandrels **102** may take any suitable configuration depending on the construction of the stiffened composite structure **12** being assembled. The example layup mandrel **102** of FIG. **3** may be described as an inner mold line layup mandrel, because the layup mandrel **102** is used to form the stiffened composite structure **12** on a radially outward surface thereof; however, outer mold line layup mandrels also may be used in some systems and methods according to the present disclosure. Again, the example of FIG. **3** is provided as an example only.

(16) Turning now to FIG. **4**, systems **100** for assembling stiffened composite structures **12** are schematically represented. Generally, in FIG. **4**, elements that are likely to be included in a given example are illustrated in solid lines, while elements that are optional to a given example are illustrated in broken lines. However, elements that are illustrated in solid lines are not essential to all examples of the present disclosure, and an element shown in solid lines may be omitted from a particular example without departing from the scope of the present disclosure. As shown, systems **100** generally comprise at least a layup mandrel **102** and a network **104** of fluid lines **116**. The network **104** of fluid lines **116** is configured to be selectively and fluidically coupled to stringer bladders **118** that are positioned relative to the layup mandrel **102**.

(17) Fluid lines **116** also may be referred to as pipes, piping, tubes, tubing, and the like. In some examples, the network **104** of fluid lines **116** is configured to be selectively and fluidically coupled to hollow stringer bladders **118** that are positioned relative to the layup mandrel **102**. In particular, in some such examples, the network **104** of fluid lines **116** is fluidically open to atmosphere **120** so that internal volumes **136** of the stringer bladders **118**, when fluidically coupled to the network **104** of fluid lines **116**, are fluidically open to atmosphere **120**. Accordingly, during a vacuum bagging operation, when a vacuum is drawn between a flexible sheet of material **154** and the layup mandrel **102**, the hollow stringer bladders **118** will not collapse.

(18) In some examples, and as schematically illustrated in FIG. **4**, the network **104** extends through, or penetrates, the layup mandrel **102**. By extending through the layup mandrel **102**, the network **104** may be open to atmosphere **120** without effecting the vacuum during a vacuum bagging operation. In particular, the outer surface of the portion of the network **104** that penetrates the layup mandrel **102** (e.g. an outer surface of a fluid line **116**) may be sealed with the layup mandrel **102** to ensure the integrity of the vacuum during a vacuum bagging operation. In this manner, there is no need for the fluid lines **116** to penetrate the flexible sheet of material **154**, thereby avoiding perforations and possible leak points that would otherwise affect the integrity of the vacuum during a vacuum bagging operation. As a result, technician time for installation of a vacuum bagging system is reduced and the need for rework associated with perforating the flexible sheet of material **154** is avoided. In addition, the flexible sheet of material **154** may be reused numerous times, unlike with conventional vacuum bagging systems.

(19) In some examples, the network **104** of fluid lines **116** may be described as comprising one or more subnetworks **119** of the fluid lines **116**. In such examples, each subnetwork **119** is configured to be selectively and fluidically coupled to more than one of the stringer bladders **118** that are positioned within the stringer forms **108** of the layup mandrel **102** so that the stringer bladders **118** are fluidically open to atmosphere **120**. In some such examples, each subnetwork **119** comprises a main line **121** and a plurality of jumper lines **122** that extend from the main line **121**. The main line **121** also may be referred to or described as a manifold **121**. Each jumper line **122** is configured to extend to a respective stringer bladder **118** that is positioned relative to the layup mandrel **102**. In some such examples, each jumper line **122** comprises a bladder-side jumper quick-connect fitting **124** that is configured to be operatively coupled to a corresponding bladder quick-connect fitting

126 of a respective bladder **118** of the plurality stringer bladders **118**. In particular, each bladder **118** may comprise a bladder vent plug **158** positioned at a terminal end of the bladder **118** and that effectively closes the hollow interior of the bladder **118** but for the connection to a vent line, such as a jumper line **122** via a quick-connect fitting assembly between the jumper line **122** and bladder vent plug **158**.

(20) Similarly, in some examples, each jumper line **122** may comprise a manifold-side jumper quick-connect fitting **125**, and the main line **121** may comprise a main-line quick-connect fitting **127** for each of the plurality of jumper lines **122** and configured to be selectively and repeatedly coupled to and uncoupled from a respective manifold-side jumper quick-connect fitting **125**.

(21) In some systems **100** that comprise subnetworks **119**, the main line **121** extends through the layup mandrel **102** and is open to atmosphere **120**. In such examples, the main line **121** may be described as a vent line **140**.

(22) In some examples, the layup mandrel **102** may comprise a face sheet **138**, and each subnetwork **119** comprises a vent line **140** that is fluidically coupled to a respective main line **121** and that extends through the face sheet **138**.

(23) With continued reference to FIG. 4, some systems **100** further comprise a leak test kit **128** for leak testing the network **104** of the fluid lines **116**, a subnetwork **119** of the fluid lines **116**, and/or the stringer bladders **118**. For example, the leak test kit **128** may comprise a gauge **130**, a plurality of jumper-line caps **132**, and an on/off valve **134**. The gauge **130** is configured to be selectively coupled to a respective subnetwork **119** and may be a pressure gauge, a flow meter, or any other suitable mechanism capable of detecting a flow of fluid or change in pressure. The jumper-line caps **132** are configured to be selectively coupled to the plurality of jumper lines **122** to isolate the respective subnetwork **119** from atmosphere **120**. The on/off valve **134** is associated with the gauge **130** and is configured to be operatively coupled to a source **133** of pressurized gas **146**.

Accordingly, the leak test kit **128** is configured to selectively couple to a subnetwork **119** to test the pressure integrity of the respective subnetwork **119**. More specifically, the ends of the jumper lines **122** of a subnetwork **119** that are configured to be operatively coupled to the stringer bladders **118** are capped with the jumper-line caps **132**, the gauge **130** and on/off valve **134** are coupled to the main line **121** or a vent line **140** of a subnetwork, and the source **133** of pressurized gas **146** is connected to the on/off valve **134**. When the on/off **134** valve is turned on, the pressurized gas **146** will pressurize the respective subnetwork **119**. Then the on/off valve **134** is closed, and the gauge **130** is monitored to detect any leaks throughout the subnetwork **119**.

(24) As schematically represented in FIG. 4, some systems **100** further comprise one or more covers **112** that are configured to be operatively coupled to the layup mandrel **102** to define a passage **114** extending along the layup mandrel **102**. In such systems **100**, the network **104** of fluid lines **116** is positioned at least partially within the passage **114** when the one or more covers **112** are operatively coupled to the layup mandrel **102**. The cover(s) **112** and the creation of the passage **114** along the open ends **110** of the stringer forms **108** may serve at least two purposes. First, the passage **114** provides a volume of space where components associated with the layup mandrel **102**, such as the network **104** of fluid lines **116**, and the assembly of stiffened composite structures **12** may be positioned and protected or otherwise shielded, such as from being engaged by a flexible sheet of material **154** of a vacuum bagging system **144**. Stated differently, the cover(s) **112** may protect the flexible sheet of material **154** from engagement with components positioned within the passage **114**, thereby avoiding undesirable wear and tear on the flexible sheet of material **154**, and resulting in a longer use life of the flexible sheet of material **154**.

(25) In some examples, as schematically represented in FIG. 4, the cover(s) **112** may comprise a smooth outer surface **141** that is suitable to interface with the flexible sheet of material **154** of a vacuum bagging system **144**. In other words, not only do the cover(s) **112** protect the flexible sheet of material **154** from the components within the passage **114**, but the cover(s) **112** themselves may be configured to avoid undesirable wear and tear of the flexible sheet of material **154**. That is, the

cover(s) **112** may not have any sharp edges or corners that are positioned to engage the flexible sheet of material **154** during a vacuum bagging operation. As a result, the flexible sheet of material **154** may be reused numerous times without replacement, unlike with conventional vacuum bagging systems.

(26) In systems **100**, whose layup mandrel **102** is an inner mold line layup mandrel with a plurality of stringer forms **108** with open ends **110**, the one or more covers **112** are configured to cover the open end **110** of each stringer form **108**, such that the passage **114** extends transverse to the plurality of spaced-apart stringer forms **108**. The cover(s) **112** and the layup mandrel **102** may have any suitable coupling mechanism or components operative to permit for the selective coupling of the cover(s) **112** to the layup mandrel **102** and for the selective uncoupling, or removal, of the cover(s) **112** from the layup mandrel **102**. Examples of coupling mechanisms or components include magnets, fasteners, clips, and the like.

(27) In some examples, the smooth outer surface **141** of the cover(s) comprises a first region **150** that is parallel to the primary skin contour **148** when the cover(s) **112** are operatively coupled to the layup mandrel **102**. Accordingly, as schematically illustrated in FIG. 4, a flexible sheet of material **154** of a vacuum bagging system **144**, when operatively installed, may smoothly transition from the first region **150** to a skin segment **38** or to an optional caul structure **156** that is placed over a skin segment **38** during a vacuum bagging operation of a stiffened composite structure **12**. In some examples, the cover(s) **112** may be described as being rigid, so as to define the passage **114** when the cover(s) **112** are operatively coupled to the layup mandrel **102**. The cover(s) **112** may be constructed of any suitable material, including plastics, fiber-reinforced plastics, metal, etc. In some examples, the cover(s) **112** may be constructed of the same material as the outer surface **106** of the layup mandrel **102**.

(28) In the example of a system **100** utilized to construct a stiffened composite structure **12** that is cylindrical or tubular, such as a barrel section **18** of a fuselage **16**, the first region **150** therefore may be cylindrical or tubular. In some examples, the smooth outer surface **141** of the cover(s) **112** also comprises a second region **152** that extends at an angle from the first region **150**. In some such examples, the second region **152** may be frustoconical.

(29) With continued reference to FIG. 4, some systems **100** further comprise a network **104** of fluid lines **116** that are positioned at least partially within the passage **114** when the cover(s) **112** are operatively coupled to the layup mandrel **102**.

(30) With continued reference to FIG. 4, systems **100** may be described in a state of use. That is a system **100** may be described as comprising a plurality of stringers **36** and a skin segment **38** operatively positioned relative to the layup mandrel **102**, and a plurality of stringer bladders **118** that are positioned within at least a subset of the plurality of stringers **36**. Further, when in use, systems **100** that comprise cover(s) **112**, the cover(s) **112** are operatively coupled to the layup mandrel **102**, and the system **100** further comprises a vacuum bagging system **144** that is operatively installed relative to the layup mandrel **102** to compact the plurality of stringers **36** and the skin segment **38** together. In particular, in some such systems **100**, the vacuum bagging system **144** comprises a flexible sheet of material **154** that extends over the one or more covers **112**. In some installations of systems **100**, the cover(s) **112** do not engage the stringers **36** and the skin segment **38**.

(31) Turning now to FIGS. 5-8, an illustrative non-exclusive example of systems **100** is illustrated in the form of a system **300**. Where appropriate, the reference numerals from the schematic illustration of FIG. 4 are used to designate corresponding parts of system **300**; however, system **300** is non-exclusive and does not limit systems **100** to the illustrated embodiment of FIGS. 5-8. That is, systems **100** may incorporate any number of the various aspects, configurations, characteristics, properties, etc. of systems **100** that are illustrated in and discussed with reference to the schematic representations of FIG. 4 and/or the embodiments of FIGS. 5-8, as well as variations thereof, without requiring the inclusion of all such aspects, configurations, characteristics, properties, etc.

For the purpose of brevity, each previously discussed component, part, portion, aspect, region, etc. or variants thereof may not be discussed, illustrated, and/or labeled again with respect to system **300**; however, it is within the scope of the present disclosure that the previously discussed features, variants, etc. may be utilized with system **300**.

(32) FIG. 5 illustrates system **300** with three covers **112** operatively coupled to a layup mandrel **102** in the form of an inner mold line layup mandrel with a plurality of spaced-apart stringer forms **108**. The example layup mandrel may be described as a segmented layup mandrel with a joint line that coincides at a stringer form **108**. For clarity, in FIG. 5, the corresponding stringers **36** and skin segment(s) **38** that would be compacted utilizing the illustrated system **300** are not shown. In particular, system **300** comprises two subnetworks **119** of fluid lines **116**. A first subnetwork **302** comprises a main line **121**, two jumper lines **122**, and two vent lines **140**. A second subnetwork **304** comprises a main line **121**, three jumper lines **122**, and two vent lines **140**.

(33) FIG. 6 illustrates system **300** but with the covers **112** removed from the layup mandrel **102** and with a leak test kit **128** installed on the subnetwork **304**. More specifically, a first gauge **130** is installed on one of the two vent lines **140**, and a second gauge **130** with an on/off valve **134** is installed on the other of the two vent lines **140** with a source **133** of pressurized gas **146** coupled thereto. The jumper lines **122** of the subnetwork **304** are connected to the respective stringer bladders **118**. Accordingly, the subnetwork **304** and the associated stringer bladders **118** are pressurized. When the on/off valve **134** is closed, the gauges **130** may be monitored to determine if any leaks are present.

(34) FIG. 7 illustrates a portion of system **300** comprising the subnetwork **304**, again with the covers **112** removed from the layup mandrel **102** and with a leak test kit **128** installed on the subnetwork **304**. However, in contrast to FIG. 6, the stringer bladders **118** are not present, and jumper-line caps **132** are operatively coupled to the jumper lines **122**. Accordingly, as illustrated, the subnetwork **304**, by itself, is pressurized, and when the on/off valve **134** is closed, the gauges **130** may be monitored to determine if any leaks are present in the subnetwork **304**.

(35) FIG. 8 again illustrates the portion of system **300** comprising the subnetwork **304**. Again, in FIG. 8, the covers **112** are moved from the layup mandrel. Two stringer bladders **118** are present and operatively coupled to corresponding jumper lines **122**. A third jumper line **122** is capped with a jumper-line cap **132**, illustrating that a system **100** may be installed and utilized in connection with fewer than the total possible number of stringer bladders **118**.

(36) FIG. 9 schematically provides a flowchart that represents illustrative, non-exclusive examples of methods **200** according to the present disclosure. In FIG. 9, some steps are illustrated in dashed boxes indicating that such steps may be optional or may correspond to an optional version of a method according to the present disclosure. That said, not all methods according to the present disclosure are required to include the steps illustrated in solid boxes. The methods and steps illustrated in FIG. 9 are not limiting and other methods and steps are within the scope of the present disclosure, including methods having greater than or fewer than the number of steps illustrated, as understood from the discussions herein.

(37) Methods **200** of utilizing a system **100** comprise operatively loading **202** stringers **36** and a skin segment **38** relative to a layup mandrel **102**, and positioning **204** stringer bladders **118** into stringer cavities of the stringers **36**. In methods **200** utilizing an inner mold line layup mandrel with a plurality of spaced-apart stringer forms **108**, the stringers **36** are positioned within the stringer forms **108**. In some methods **200**, the stringers **36** and the stringer bladders **118** may be kitted away from the layup mandrel **102** and loaded on the layup mandrel **102** together. In other methods **200**, the stringers **36** may be positioned in the stringer forms **108** of the layup mandrel **102** before the stringer bladders **118** are positioned within the cavities of the stringers **36**.

(38) Methods **200** further comprise fluidically coupling **206** the stringer bladders **118** to a network **104** of fluid lines **116** so that the internal volumes **136** of the stringer bladders **118** are fluidically open to atmosphere **120**, and operatively installing **212** a vacuum bagging system **144** relative to

the layup mandrel **102**, the stringers **36**, and the skin segment **38** that loaded on the layup mandrel **102**. Following the operatively installing **212** the vacuum bagging system **144**, methods **200** comprise drawing **214** a vacuum on the stringers **36** and the skin segment **38** while maintaining atmospheric pressure of the stringer bladders **118** via the network **104** of fluid lines **116**. Methods **200** further comprise concurrently with the drawing **214** the vacuum, curing **216** the stringers **36** and the skin segment **38**.

(39) In methods **200** that utilize a system **100** that comprises one more covers **112**, such methods **200** also may comprise, following the fluidically coupling **206**, operatively coupling **208** one or more covers **112** to the layup mandrel **102** to define a passage **114**.

(40) As schematically represented in FIG. **9**, some methods **200** further comprise, following the curing **216**, uninstalling **218** the vacuum bagging system **144** from the layup mandrel **102**, and following the uninstalling **218**, removing **220** the one or more covers **112** from the layup mandrel **102**. In addition in such examples of methods **200**, following the removing **220** the one or more covers **112**, methods **200** further comprise disconnecting **222** the stringer bladders **118** from the network **104** of fluid lines **116**, and removing **224** the stringer bladders **118** from the stringers **36**. Following the uninstalling **218**, such methods **200** also comprise removing **226** the stringers **36** and the skin segment **38** from the layup mandrel **102**.

(41) Some methods **200** further comprise prior to the fluidically coupling **206** the stringer bladders **118** to the network **104** of fluid lines **116**, capping **228** the plurality of jumper lines **122** of a respective subnetwork **119**, operatively coupling **230** a gauge **130** to the respective subnetwork **119**, and pressurizing **232** the respective subnetwork **119** to leak test the respective subnetwork **119**.

(42) Some methods **200** further comprise, following the fluidically coupling **206** the stringer bladders **118** to the network **104** of fluid lines **116**, operatively coupling **234** a gauge **130** to the respective subnetwork **119**, and pressurizing **236** the respective subnetwork **119** and the stringer bladders **118** fluidically coupled to the respective subnetwork **119** to leak test the stringer bladders **118** that are fluidically coupled to the respective subnetwork **119**.

(43) Illustrative, non-exclusive examples of inventive subject matter according to the present disclosure are described in the following enumerated paragraphs:

(44) A. A system (**100**) for assembling stiffened composite structures (**12**), the system (**100**) comprising: a layup mandrel (**102**) comprising an outer surface (**106**); and a network (**104**) of fluid lines (**116**), wherein the network (**104**) of fluid lines (**116**) is configured to be selectively and fluidically coupled to stringer bladders (**118**) positioned relative to the layup mandrel (**102**).

(45) A1. The system (**100**) of paragraph A, wherein the network (**104**) of fluid lines (**116**) is fluidically open to atmosphere (**120**) so that internal volumes (**136**) of the stringer bladders (**118**), when fluidically coupled to the network (**104**) of fluid lines (**116**), are fluidically open to atmosphere (**120**).

(46) A2. The system (**100**) of any of paragraphs A-A1, wherein the network (**104**) of fluid lines (**116**) extends through the layup mandrel (**102**).

(47) A3. The system (**100**) of any of paragraphs A-A2, wherein the network (**104**) of fluid lines (**116**) comprises one or more subnetworks (**119**) of the fluid lines (**116**), wherein each subnetwork (**119**) of the one or more subnetworks (**119**) is configured to be selectively and fluidically coupled to more than one of the stringer bladders (**118**) so that the more than one of the stringer bladders (**118**) are fluidically open to atmosphere (**120**).

(48) A3.1. The system (**100**) of paragraph A3, wherein each subnetwork (**119**) of the one or more subnetworks (**119**) comprises a main line (**121**) and a plurality of jumper lines (**122**) extending from the main line (**121**), wherein each jumper line (**122**) of the plurality of jumper lines (**122**) is configured to be operatively coupled to a respective stringer bladder (**118**) of the stringer bladders (**118**).

(49) A3.1.1. The system (**100**) of paragraph A3.1, wherein each jumper line (**122**) of the plurality of jumper lines (**122**) comprises a bladder-side jumper quick-connect fitting (**124**) configured to be

operatively coupled to a corresponding bladder quick-connect fitting (126) of the respective bladder (118) of the stringer bladders (118).

(50) A3.1.2. The system (100) of any of paragraphs A3.1-A3.1.1, wherein each jumper line (122) of the plurality of jumper lines (122) comprises a manifold-side jumper quick-connect fitting (125), and wherein the main line (121) comprises a main-line quick-connect fitting (127) for each jumper line (122) of the plurality of jumper lines (122).

(51) A3.1.3. The system (100) of any of paragraphs A3.1-A3.1.2, wherein the main line (121) of each subnetwork (119) of the one or more subnetworks (119) extends through the layup mandrel (102).

(52) A3.1.4. The system (100) of any of paragraphs A3.1-A3.1.3, wherein the layup mandrel (102) comprises a face sheet (138), wherein each subnetwork (119) of the one or more subnetworks (119) further comprises at least one vent line (140), and wherein each vent line (140) is fluidically coupled to a respective main line (121) through the face sheet (138).

(53) A3.2. The system (100) of any of paragraphs A3-A3.1.4, further comprising a leak test kit (128) comprising: a gauge (130) configured to be selectively coupled to a respective subnetwork (119) of the one or more subnetworks (119); a plurality of jumper-line caps (132) configured to be selectively coupled to the plurality of jumper lines (122); and an on/off valve (134) associated with the gauge (130) and configured to be operatively coupled to a source (133) of pressurized gas (146); wherein the leak test kit (128) is configured to selectively pressurize the respective subnetwork (119) of the one or more subnetworks (119) of the fluid lines (116) to test pressure integrity of the respective subnetwork (119).

(54) A4. The system (100) of any of paragraphs A-A3.2, further comprising: one or more covers (112) configured to be operatively coupled to the layup mandrel (102) to define a passage (114) extending along the layup mandrel (102); wherein the network (104) of fluid lines (116) is positioned at least partially within the passage (114) when the one or more covers (112) are operatively coupled to the layup mandrel (102), and wherein the network (104) of fluid lines (116) is configured to be selectively and fluidically coupled to the stringer bladders (118) when positioned relative to the layup mandrel (102).

(55) A4.1. The system (100) of paragraph A4, wherein the one or more covers (112) comprise a smooth outer surface (141) suitable to interface with a flexible sheet of material (154) of a vacuum bagging system (144).

(56) A4.2. The system of any of paragraphs A4-A4.1, wherein the outer surface (106) of the layup mandrel (102) defines a plurality of spaced-apart stringer forms (108) extending longitudinally along the outer surface (106), wherein each stringer form (108) of the plurality of spaced-apart stringer forms (108) comprises an open end (110); wherein when the one or more covers (112) are operatively coupled to the layup mandrel (102), the one or more covers (112) cover the open end (110) of each stringer form (108) of the plurality of spaced-apart stringer forms (108), and the passage (114) extends transverse to the plurality of spaced-apart stringer forms (108); and wherein the network (104) of fluid lines (116) is configured to be selectively and fluidically coupled to the stringer bladders (118) when positioned within the plurality of spaced-apart stringer forms (108).

(57) A4.2.1. The system (100) of paragraph A4.2 when depending from paragraph A4.1, wherein the outer surface (106) of the layup mandrel (102) defines a primary skin contour (148), and wherein the smooth outer surface (141) of the one or more covers (112) comprises a first region (150) that is parallel to the primary skin contour (148) when operatively coupled to the layup mandrel (102).

(58) A4.2.1.1. The system (100) of paragraph A4.2.1, wherein the smooth outer surface (141) of the one or more covers (112) comprises a second region (152) that extends at an angle from the first region (150).

(59) A4.2.1.1.1. The system (100) of paragraph A4.2.1.1, wherein the first region (150) is cylindrical, and wherein the second region (152) is frustoconical.

(60) A5. The system (100) of any of paragraphs A-A4.2.1.1.1, further comprising: a plurality of stringers (36) and a skin segment (38) operatively positioned relative to the layup mandrel (102); and a plurality of stringer bladders (118) positioned within at least a subset of the plurality of stringers (36).

(61) A5.1. The system (100) of paragraph A5 when depending from paragraph A4, wherein the one or more covers (112) are operatively coupled to the layup mandrel (102), and wherein the system (100) further comprises a/the vacuum bagging system (144) operatively installed relative to the layup mandrel (102) to compact the plurality of stringers (36) and the skin segment (38).

(62) A5.1.1. The system (100) of paragraph A5.1, wherein the vacuum bagging system (144) comprises a/the flexible sheet of material (154) extending over the one or more covers (112).

(63) A5.1.1.1. The system (100) of paragraph A5.1.1, wherein the fluid lines (116) do not penetrate the flexible sheet of material (154).

(64) A5.1.2. The system (100) of any of paragraphs A5.1-A5.1.1.1, wherein the one or more covers (112) do not engage the plurality of stringers (36) and the skin segment (38).

(65) A6. The system (100) of any of paragraphs A-A5.1.2, wherein the layup mandrel (102) corresponds to at least a portion of a fuselage (16) of an aircraft (14).

(66) A7. The use of the system (100) of any of paragraphs A-A6 to assemble a/the stiffened composite structure (12).

(67) B. A method (200) of utilizing the system (100) of any of paragraphs A-A7, the method (200) comprising: operatively loading (202) stringers (36) and a/the skin segment (38) relative to the layup mandrel (102); positioning (204) the stringer bladders (118) into stringer cavities of the stringers (36); fluidically coupling (206) the stringer bladders (118) to the network (104) of fluid lines (116) so that the internal volumes (136) of the stringer bladders (118) are fluidically open to atmosphere (120); operatively installing (212) a/the vacuum bagging system (144) relative to the layup mandrel (102), the stringers (36), and the skin segment (38); following the operatively installing (212) the vacuum bagging system (144), drawing (214) a vacuum on the stringers (36) and the skin segment (38) while maintaining atmospheric pressure of the stringer bladders (118) via the network (104) of fluid lines (116); and concurrently with the drawing (214) the vacuum, curing (216) the stringers (36) and the skin segment (38).

(68) B1. The method (200) of paragraph B when depending from paragraph A4, further comprising:

(69) following the fluidically coupling (206), operatively coupling (208) the one or more covers (112) to the layup mandrel (102) to define the passage (114).

(70) B1.1. The method (200) of paragraph B1, further comprising: following the curing (216), uninstalling (218) the vacuum bagging system (144) from the layup mandrel (102); following the uninstalling (218), removing (220) the one or more covers (112) from the layup mandrel (102); following the removing (220) the one or more covers (112): disconnecting (222) the stringer bladders (118) from the network (104) of fluid lines (116); and removing (224) the stringer bladders (118) from the stringers (36); and following the uninstalling (218), removing (226) the stringers (36) and the skin segment (38) from the layup mandrel (102).

(71) B2. The method (200) of any of paragraphs B-B1.1 when depending from paragraph A3.1, further comprising, prior to the fluidically coupling (206) the stringer bladders (118) to the network (104) of fluid lines (116): capping (228) the plurality of jumper lines (122) of a respective subnetwork (119); operatively coupling (230) a gauge (130) to the respective subnetwork (119); and pressurizing (232) the respective subnetwork (119) to leak test the respective subnetwork (119).

(72) B3. The method (200) of any of paragraphs B-B2 when depending from paragraph A3.1, further comprising, following the fluidically coupling (206) the stringer bladders (118) to the network (104) of fluid lines (116): operatively coupling (234) a/the gauge (130) to the respective subnetwork (119); and pressurizing (236) the respective subnetwork (119) and the stringer bladders

(118) fluidically coupled to the respective subnetwork (119) to leak test the stringer bladders (118) fluidically coupled to the respective subnetwork (119).

(73) B4. The method (200) of any of paragraphs B-B3, wherein the operatively installing (212) does not comprise penetrating a/the flexible sheet of material (154) of the vacuum bagging system (144) with the fluid lines (116). As used herein, the terms “adapted” and “configured” mean that the element, component, or other subject matter is designed and/or intended to perform a given function. Thus, the use of the terms “adapted” and “configured” should not be construed to mean that a given element, component, or other subject matter is simply “capable of” performing a given function but that the element, component, and/or other subject matter is specifically selected, created, implemented, utilized, programmed, and/or designed for the purpose of performing the function. It is also within the scope of the present disclosure that elements, components, and/or other recited subject matter that is recited as being adapted to perform a particular function may additionally or alternatively be described as being configured to perform that function, and vice versa. Similarly, subject matter that is recited as being configured to perform a particular function may additionally or alternatively be described as being operative to perform that function.

(74) As used herein, the term “and/or” placed between a first entity and a second entity means one of (1) the first entity, (2) the second entity, and (3) the first entity and the second entity. Multiple entries listed with “and/or” should be construed in the same manner, i.e., “one or more” of the entities so conjoined. Other entities optionally may be present other than the entities specifically identified by the “and/or” clause, whether related or unrelated to those entities specifically identified. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising,” may refer, in one example, to A only (optionally including entities other than B); in another example, to B only (optionally including entities other than A); in yet another example, to both A and B (optionally including other entities). These entities may refer to elements, actions, structures, steps, operations, values, and the like.

(75) The various disclosed elements of apparatuses and steps of methods disclosed herein are not required to all apparatuses and methods according to the present disclosure, and the present disclosure includes all novel and non-obvious combinations and subcombinations of the various elements and steps disclosed herein. Moreover, one or more of the various elements and steps disclosed herein may define independent inventive subject matter that is separate and apart from the whole of a disclosed apparatus or method. Accordingly, such inventive subject matter is not required to be associated with the specific apparatuses and methods that are expressly disclosed herein, and such inventive subject matter may find utility in apparatuses and/or methods that are not expressly disclosed herein.

Claims

1. A system for assembling stiffened composite structures, the system comprising: a layup mandrel comprising an outer surface; a network of fluid lines, wherein the network of fluid lines is configured to be selectively and fluidically coupled to stringer bladders positioned relative to the layup mandrel, and wherein the network of fluid lines extends through the layup mandrel; and one or more covers configured to be operatively coupled to the layup mandrel to define a passage extending along the layup mandrel; wherein the network of fluid lines is positioned at least partially within the passage when the one or more covers are operatively coupled to the layup mandrel, and wherein the network of fluid lines is configured to be selectively and fluidically coupled to the stringer bladders when positioned relative to the layup mandrel.
2. The system of claim 1, wherein the network of fluid lines is fluidically open to atmosphere so that internal volumes of the stringer bladders, when fluidically coupled to the network of fluid lines, are fluidically open to atmosphere.
3. The system of claim 1, wherein the network of fluid lines comprises one or more subnetworks of

the fluid lines, wherein each subnetwork of the one or more subnetworks is configured to be selectively and fluidically coupled to more than one of the stringer bladders so that the more than one of the stringer bladders are fluidically open to atmosphere.

4. The system of claim 3, wherein each subnetwork of the one or more subnetworks comprises a main line and a plurality of jumper lines extending from the main line, wherein each jumper line of the plurality of jumper lines is configured to be operatively coupled to a respective stringer bladder of the stringer bladders.

5. The system of claim 4, wherein each jumper line of the plurality of jumper lines comprises a bladder-side jumper quick-connect fitting configured to be operatively coupled to a corresponding bladder quick-connect fitting of the respective stringer bladder of the stringer bladders.

6. The system of claim 4, wherein each jumper line of the plurality of jumper lines comprises a manifold-side jumper quick-connect fitting, and wherein the main line comprises a main-line quick-connect fitting for each jumper line of the plurality of jumper lines.

7. The system of claim 4, wherein the main line of each subnetwork of the one or more subnetworks extends through the layup mandrel.

8. The system of claim 4, wherein the layup mandrel comprises a face sheet, wherein each subnetwork of the one or more subnetworks further comprises at least one vent line, and wherein each vent line is fluidically coupled to a respective main line through the face sheet.

9. The system of claim 3, further comprising a leak test kit comprising: a gauge configured to be selectively coupled to a respective subnetwork of the one or more subnetworks; a plurality of jumper-line caps configured to be selectively coupled to the plurality of jumper lines; and an on/off valve associated with the gauge and configured to be operatively coupled to a source of pressurized gas; wherein the leak test kit is configured to selectively pressurize the respective subnetwork of the one or more subnetworks of the fluid lines to test pressure integrity of the respective subnetwork.

10. The system of claim 1, wherein the one or more covers comprise a smooth outer surface suitable to interface with a flexible sheet of material of a vacuum bagging system.

11. The system of claim 1, wherein the outer surface of the layup mandrel defines a plurality of spaced-apart stringer forms extending longitudinally along the outer surface, wherein each stringer form of the plurality of spaced-apart stringer forms comprises an open end; wherein when the one or more covers are operatively coupled to the layup mandrel, the one or more covers cover the open end of each stringer form of the plurality of spaced-apart stringer forms, and the passage extends transverse to the plurality of spaced-apart stringer forms; and wherein the network of fluid lines is configured to be selectively and fluidically coupled to the stringer bladders when positioned within the plurality of spaced-apart stringer forms.

12. The system of claim 11, wherein the one or more covers comprise a smooth outer surface suitable to interface with a flexible sheet of material of a vacuum bagging system, wherein the outer surface of the layup mandrel defines a primary skin contour, and wherein the smooth outer surface of the one or more covers comprises a first region that is parallel to the primary skin contour when operatively coupled to the layup mandrel.

13. The system of claim 12, wherein the smooth outer surface of the one or more covers comprises a second region that extends at an angle from the first region.

14. The system of claim 13, wherein the first region is cylindrical, and wherein the second region is frustoconical.

15. The system of claim 11, further comprising: a plurality of stringers positioned within at least a subset of the plurality of spaced-apart stringer forms; a plurality of stringer bladders positioned within at least a subset of the plurality of stringers positioned within at least the subset of the plurality of spaced-apart stringer forms; and a skin segment positioned over the plurality of stringers and the plurality of stringer bladders.

16. The system of claim 15, wherein the one or more covers are operatively coupled to the layup

mandrel, and wherein the system further comprises a vacuum bagging system operatively installed relative to the layup mandrel to compact the plurality of stringers and the skin segment.

17. The system of claim 16, wherein the vacuum bagging system comprises a flexible sheet of material extending over the one or more covers, and wherein the fluid lines do not penetrate the flexible sheet of material.

18. The system of claim 16, wherein the one or more covers do not engage the plurality of stringers and the skin segment.

19. The system of claim 11, wherein the layup mandrel corresponds to at least a portion of a fuselage of an aircraft, and wherein the plurality of spaced-apart stringer forms correspond to stringers of the fuselage.

20. A method of utilizing the system of claim 1, the method comprising: operatively loading stringers and a skin segment relative to the layup mandrel; positioning the stringer bladders into stringer cavities of the stringers; fluidically coupling the stringer bladders to the network of fluid lines so that internal volumes of the stringer bladders are fluidically open to atmosphere; operatively installing a vacuum bagging system relative to the layup mandrel, the stringers, and the skin segment; following the operatively installing the vacuum bagging system, drawing a vacuum on the stringers and the skin segment while maintaining atmospheric pressure of the stringer bladders via the network of fluid lines; and concurrently with the drawing the vacuum, curing the stringers and the skin segment.

21. A system for assembling stiffened composite structures, the system comprising: a layup mandrel comprising an outer surface; a network of fluid lines, wherein the network of fluid lines is configured to be selectively and fluidically coupled to stringer bladders positioned relative to the layup mandrel; and one or more covers configured to be operatively coupled to the layup mandrel to define a passage extending along the layup mandrel; wherein the network of fluid lines is positioned at least partially within the passage when the one or more covers are operatively coupled to the layup mandrel, and wherein the network of fluid lines is configured to be selectively and fluidically coupled to the stringer bladders when positioned relative to the layup mandrel.

22. A system for assembling stiffened composite structures, the system comprising: a layup mandrel comprising an outer surface; and a network of fluid lines, wherein the network of fluid lines is configured to be selectively and fluidically coupled to stringer bladders positioned relative to the layup mandrel, and wherein the network of fluid lines extends through the layup mandrel; wherein the network of fluid lines comprises one or more subnetworks of the fluid lines, wherein each subnetwork of the one or more subnetworks is configured to be selectively and fluidically coupled to more than one of the stringer bladders so that the more than one of the stringer bladders are fluidically open to atmosphere; wherein each subnetwork of the one or more subnetworks comprises a main line and a plurality of jumper lines extending from the main line, wherein each jumper line of the plurality of jumper lines is configured to be operatively coupled to a respective stringer bladder of the stringer bladders; and wherein the layup mandrel comprises a face sheet, wherein each subnetwork of the one or more subnetworks further comprises at least one vent line, and wherein each vent line is fluidically coupled to a respective main line through the face sheet.

23. A system for assembling stiffened composite structures, the system comprising: a layup mandrel comprising an outer surface; a network of fluid lines, wherein the network of fluid lines is configured to be selectively and fluidically coupled to stringer bladders positioned relative to the layup mandrel, wherein the network of fluid lines extends through the layup mandrel, wherein the network of fluid lines comprises one or more subnetworks of the fluid lines, and wherein each subnetwork of the one or more subnetworks is configured to be selectively and fluidically coupled to more than one of the stringer bladders so that the more than one of the stringer bladders are fluidically open to atmosphere; and a leak test kit comprising: a gauge configured to be selectively coupled to a respective subnetwork of the one or more subnetworks; a plurality of jumper-line caps configured to be selectively coupled to the plurality of jumper lines; and an on/off valve associated

with the gauge and configured to be operatively coupled to a source of pressurized gas; wherein the leak test kit is configured to selectively pressurize the respective subnetwork of the one or more subnetworks of the fluid lines to test pressure integrity of the respective subnetwork.
